

Financial Institutions and System

Week 13: ESG and Sustainable Finance: The Role of Financial Institutions in Promoting Sustainability.

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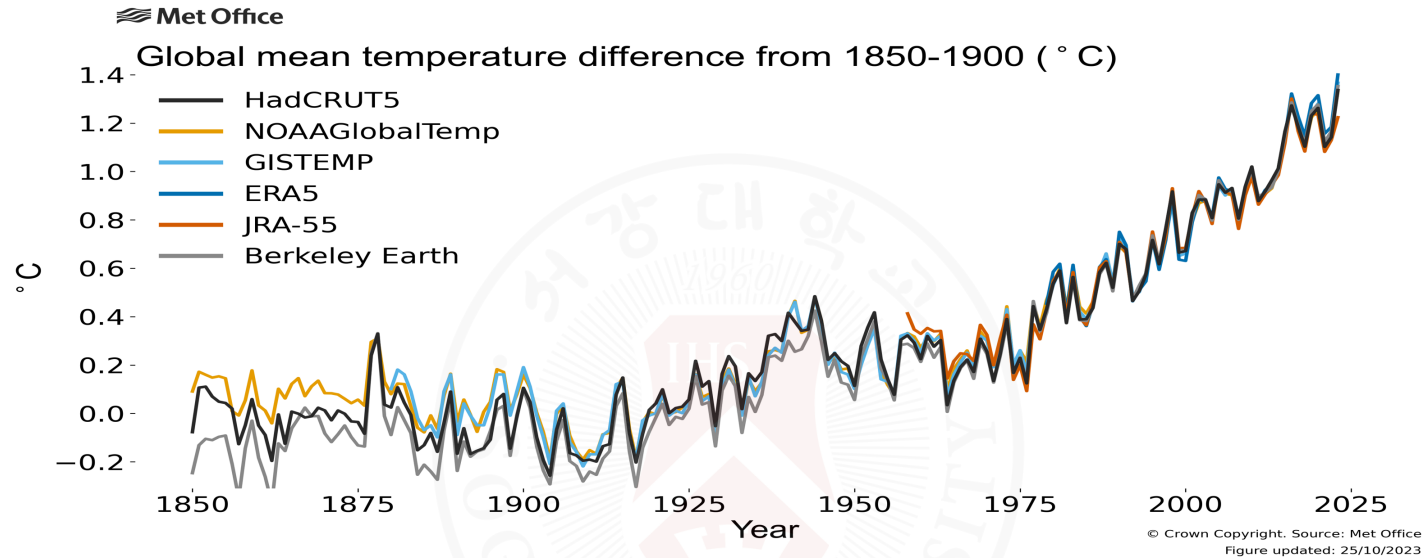
May 14, 2026

Agenda

1. ESG and Sustainable Finance: The Role of Financial Institutions in Promoting Sustainability
2. Class Activity



Climate Change: It's Getting Warmer

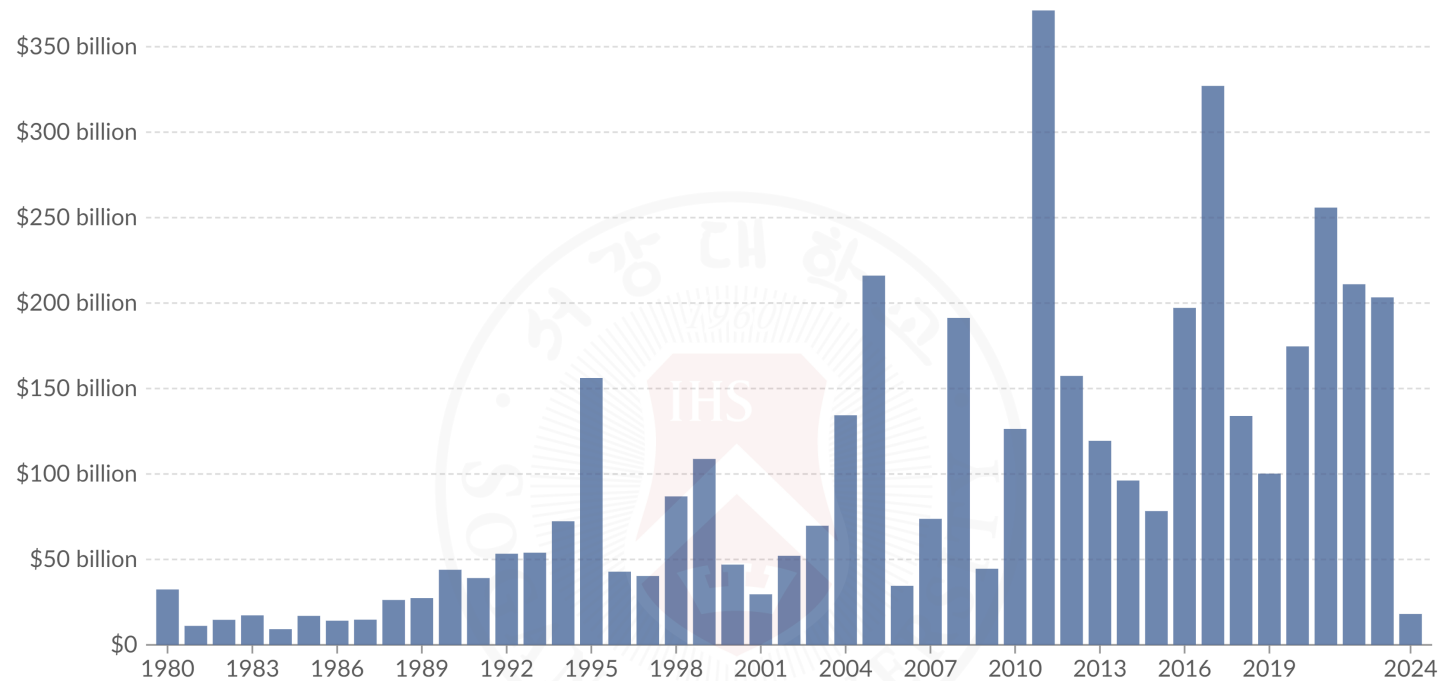


(Source: Met Office Climate Dashboard)

- Global average temperatures continue to rise relative to pre-industrial levels.
- Climate change is no longer a distant risk — it's a present reality.

2023 and 2024 were among the warmest years on record globally.

The Cost of Climate Disasters



(Source: Our World in Data based on EM-DAT)

- Disasters impose significant and rising economic costs relative to GDP.
- Wildfires, floods, and storms cause both direct and indirect damage.

Economic losses are unevenly distributed, with emerging economies often more vulnerable.

Financial System's Contribution

More banks pulled back from climate change mitigation policies, presumably contributing to this increase in fossil fuel financing.

The **65** banks in this year's report have committed **\$7.9 T USD** in fossil fuel financing since 2016, when the Paris Agreement went into effect.

Since 2021, global banks have financed **\$1.6 T USD** to fossil fuel expansion companies.

(Source: Banking on Climate Chaos Report 2025)

- Major global banks have continued to provide trillions of dollars in fossil-fuel financing since the Paris Agreement.
- \$7.9 trillion invested since the Paris Agreement.

The background of the slide features a large, faint, circular seal of Gwangju University. The seal contains the university's name in Korean and English, the year 1960, and a central shield with a crown and the letters 'IHS'.

1. ESG and Sustainable Finance: The Role of Financial Institutions in Promoting Sustainability

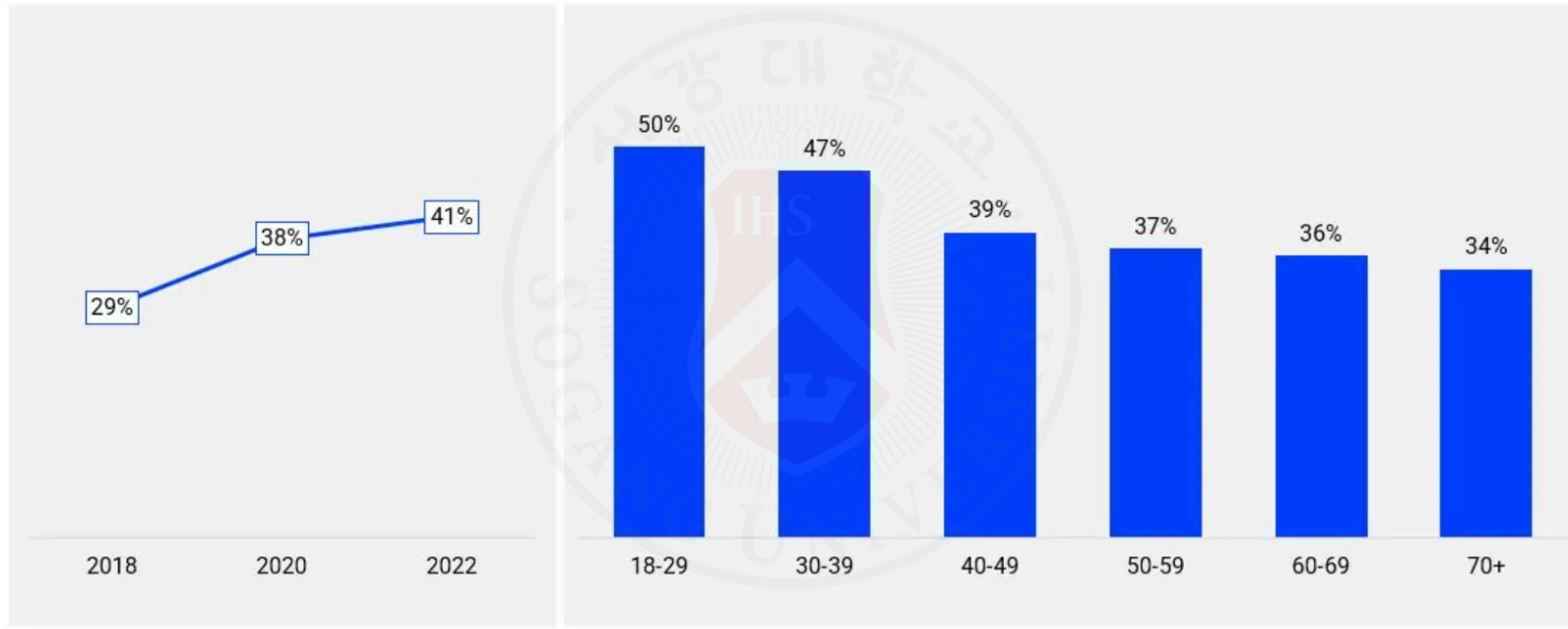
Introduction: What is ESG and Sustainable Finance?

- **ESG:** Environmental, Social, and Governance criteria used to evaluate corporate behavior
 - ESG is one framework used within broader sustainable-finance practices.
- **Sustainable Finance:** Financial activities that consider ESG factors to support long-term economic and societal well-being
 - "Sustainable finance integrates environmental, social, and governance considerations into financial decision-making, aiming to balance economic advancement with ecological preservation and social justice."

Climate change = stress

Climate change is a source of stress

% of US households that agree



RFI GLOBAL

Source: RFI Global, MacroMonitor US Households Study

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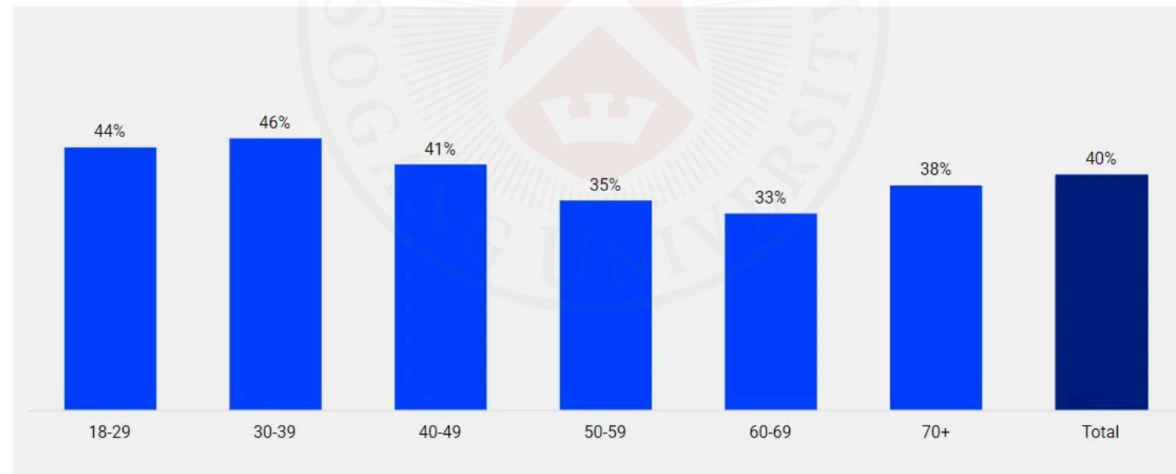
Climate risk can affect both borrowers and financial institutions directly.

Why ESG Matters in Finance

- Reputational risk reduction (e.g., scandals, greenwashing)
- Some studies suggest ESG may improve long-term resilience and risk management
- Rising regulatory demands (EU taxonomy, SFDR)
- Societal expectation: 40% of U.S. households would sacrifice some return for ESG-aligned investments

I would be willing to sacrifice some return to invest in companies that reflect my values

% of US households that agree



RFI GLOBAL

Source: RFI Global, MacroMonitor US Households Study

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ESG Principles Breakdown

Environmental

- Climate risk management
- Green investments
- Sustainable resource use

Social

- Financial inclusion
- Labor rights
- Community development

Governance

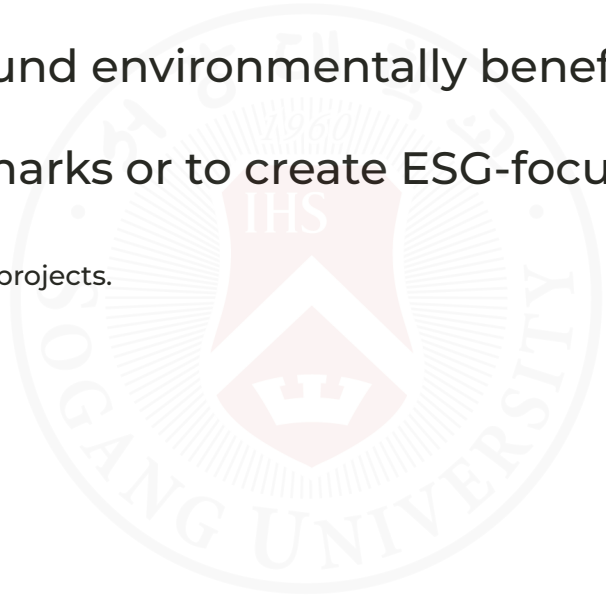
- Transparency and disclosures
- Anti-corruption
- Board diversity
- Executive compensation alignment
- Shareholder rights



ESG Integration: Methods Used

- **ESG Ratings:** Evaluate ESG performance across firms
- **Impact Investing:** Intentional investments generating measurable societal benefits
- **Green/blue Bonds & Loans:** Fund environmentally beneficial projects
- **ESG Indexes:** Used as benchmarks or to create ESG-focused portfolios

Blue bonds finance ocean and marine sustainability projects.



ESG Metrics in Practice

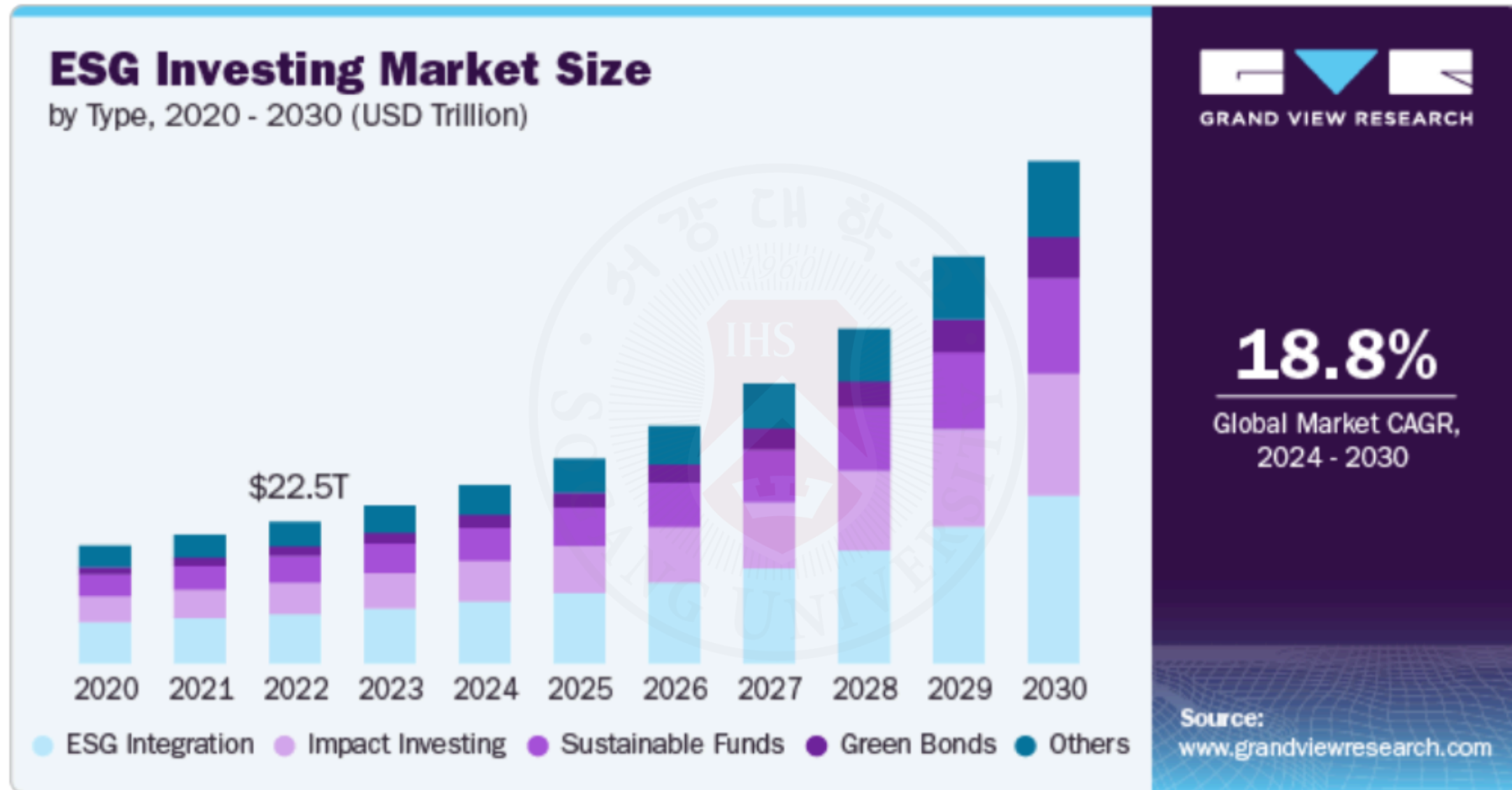
ENVIRONMENTAL	SOCIAL	GOVERNANCE
GHG emissions	Employee Diversity & Inclusion	Cybersecurity
•Amount of Scope 1, 2, and 3 emissions	•Percentage of gender / racial or ethnic group representation for management and employees	•Number of data breaches, percentage breaches involving personally identifiable information, number of users affected
Energy management	Labor practices	Business ethics
•Total energy consumed, percentage grid electricity, percentage renewable electricity	•Percentage of active workforce covered under collective bargaining agreements	•Amount of net revenue in countries that have the twenty lowest rankings in Transparency International's Corruption Perception Index
Waste management	Employee health and safety	Business Model Resilience
•Total weight of waste diverted from disposal	•Total recordable incident rate	•Amount of material recycled, composted, and processed as waste energy
Water management	Pay equity	Remuneration
•Total water consumption from all areas with water stress	•Ratios of standard entry level wage by gender compared to local minimum wage	•Annual total compensation ratio of CEO to median for all employees

- ESG integration varies across sectors
- Disclosure standards: GRI, SASB, TCFD
- Governance typically has clearer measurement frameworks than social or environmental dimensions

Source: **Novisto**

ESG ratings from different providers often disagree substantially, creating challenges for investors and companies alike.

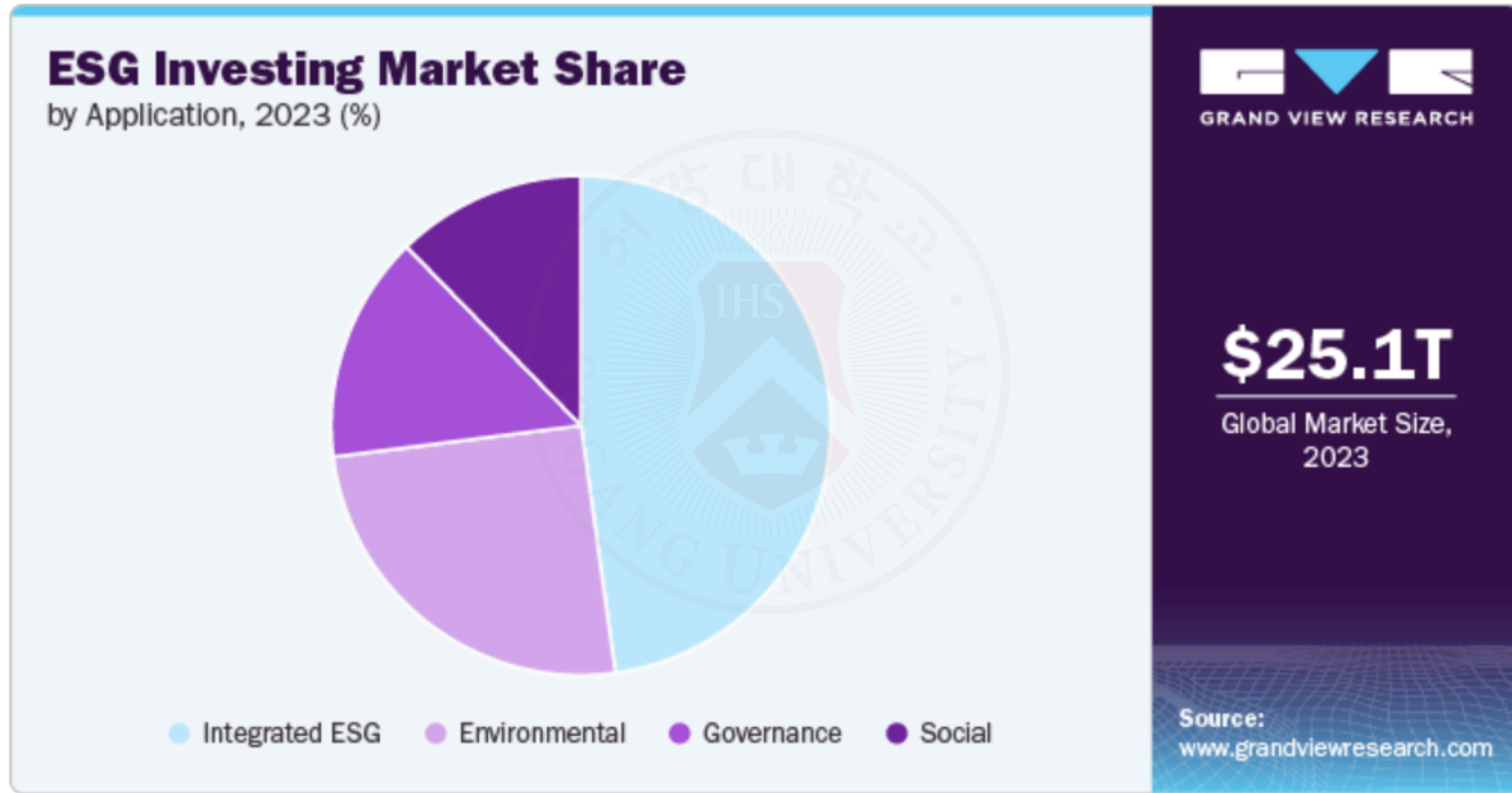
ESG Investing Market Overview



Source: **Grand View Research**

Global ESG market estimates vary significantly depending on methodology and classification.

ESG Investing Market Overview (cont.)

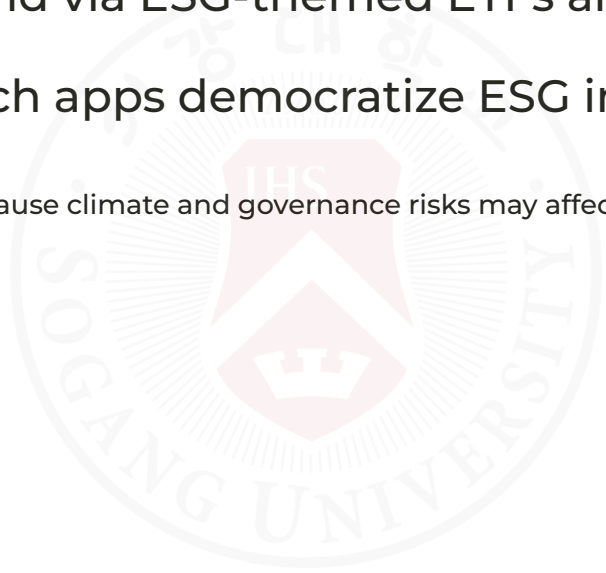


Source: Grand View Research

Investor Landscape in ESG

- **Institutional Investors (55.7%)** dominate ESG investing due to risk management and fiduciary duty
- **Retail Investors** gaining ground via ESG-themed ETFs and robo-advisors
 - | Online platforms and fintech apps democratize ESG investing for individuals

Institutional investors often integrate ESG partly because climate and governance risks may affect long-term portfolio value.



Financial Institutions as ESG Catalysts

Banking Sector

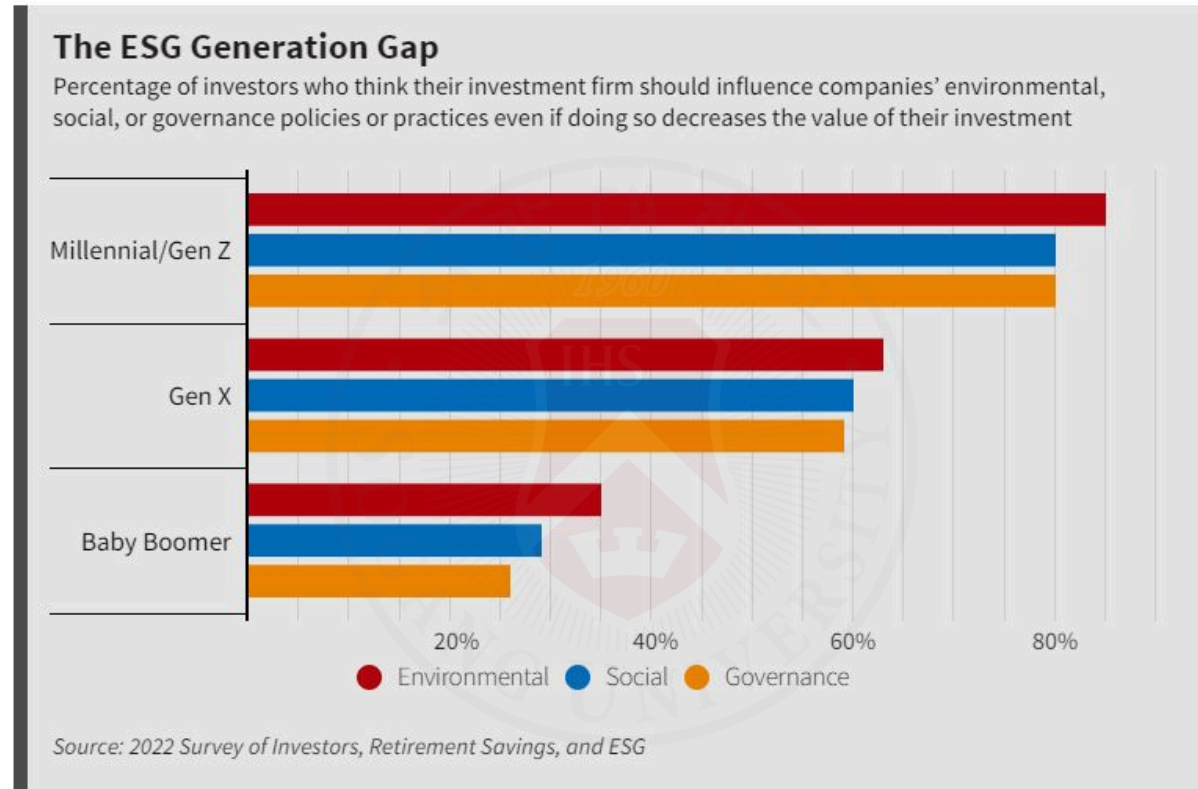
- Green deposits
- Sustainable infrastructure financing
- Transition financing for carbon-intensive sectors

Investment Sector

- ESG ETFs and mutual funds
- Stewardship and shareholder engagement

Large global financial institutions increasingly offer ESG-linked products and transition-finance initiatives

Consumer Expectations and Sentiment



- Younger investors (<40) show stronger preference for ESG
- ESG values shape client loyalty and product design
- Transparency and social impact are now competitive advantages

Global ESG Policy and Practice Trends

- Rise of climate-related financial risk disclosures
- EU Sustainable Finance Disclosure Regulation (SFDR)
- Sovereign green bonds (India, EU, others)
- Global ESG market estimated at \$35 trillion+ assets under management
- ISSB (IFRS S1 and S2) standards are becoming a major global framework for sustainability and climate disclosures
- Importance of financing transition of carbon-intensive sectors

ESG adoption is now seen as essential to financial and economic stability, though some countries and U.S. states have also seen anti-ESG political backlash

Case: ESG During and After COVID-19

During the crisis (2020–2021)

- Short-term survival overrode long-term ESG commitments
- Many firms paused sustainability reporting and green investments
- Supply chain disruptions exposed social governance gaps

Post-pandemic rebound (2022–present)

- Resilience and stakeholder capitalism moved to center stage
- ESG AUM surged as investors reassessed systemic risks
- "S" factor gained prominence: labor rights, health equity, digital access

Key lessons for financial institutions

- ESG frameworks must be **crisis-resilient**, not crisis-contingent
- Climate risk and pandemic risk share structural similarities
- Firms with strong ESG foundations showed faster recovery

"COVID-19 did not derail ESG — it stress-tested it and revealed its importance."

ESG in Central Banks and Regulation

- **ECB:** ECB increasingly integrates climate-related financial risk into supervision, stress testing, and collateral frameworks
- **RBI (India):** Draft Disclosure Framework on Climate-related Risks (2024), Green Bond issuance
- **NGFS:** Global platform for green central banking
- **ESG Reporting:** Alignment with GRI, TCFD, EU CSRD

Regulatory clarity drives better ESG integration and accountability ESG in financial regulation is becoming mandatory, not optional Climate stress testing is becoming increasingly common among central banks and regulators

Challenges and Criticism of ESG

- ESG ratings often differ across providers
 - Risk of greenwashing and misleading disclosures
 - Difficulty measuring social and environmental impact
 - Debate over whether ESG improves returns
 - Political backlash against ESG in some countries
 - Tension between sustainability goals and energy security
- | ESG remains an evolving and contested framework.

Summary & Discussion

- ESG is reshaping how banks and asset managers operate
- Regulation, data, and consumer pressure drive ESG adoption
- Financial institutions play a central role in steering capital toward sustainability
- ESG is not just risk mitigation—it's also opportunity creation
- ESG implementation remains controversial because of measurement challenges, greenwashing concerns, and political disagreements

Q&A and Reflection

- What is the biggest challenge to ESG integration?
- Which ESG factor (E, S, G) is most critical in your context?
- How should regulators ensure transparency without stifling innovation?



Used resources

- Mishkin, F. S. (2021). *Economics of Money, Banking, and Financial Markets*, 12th Edition
- Zairis (2024) *Sustainable Finance and ESG Importance*
- RFI Global MacroMonitor Reports
- Dan Byrne, *Guide to ESG in the Finance Sector*
- Grand View Research ESG Market Reports



The background of the slide features a large, faint, circular seal of Gangneung University. The seal contains the university's name in Korean and English, the year 1960, and a central shield with the letters 'IHS' and a crown.

2. Class Activity

Looking Ahead: Course Roadmap

We are entering the final phase of the course. Here's what's coming:

Session Schedule & Topics

- **(May 28) FinTech Revolution**
 - Guest Speaker (in-person)
 - How technology is reshaping financial services
- **(June 4) AI & Machine Learning in Finance**
 - Guest Speaker (online)
 - Final Exam Review Session
- **(June 11) Student Project Presentations**
 - Showcase your applied research and insights
- **(June 18) Final Examination**
 - Comprehensive assessment of course material



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Any Questions?

Thank you for your attention and participation!